Reference and annotation comparison

Arabidopsis TAIR10 and Col-CEN; fixed inputs and explicit coordinate mapping

October 16, 2026

Abstract

At 1 million ATAC pairs, reciprocal peak Jaccard is 0.8984 and 0.9639 between TAIR10 and Col-CEN for two replicates. A completed local 5-million-pair assessment gives 0.9547 and 0.9571 among eligible mappings. Annotation alone changes TSS scores on identical aligned reads, and mapping exclusions limit every cross-assembly claim. We would choose reference and annotation explicitly rather than rank them by peak count or TSS enrichment.

Setup

Data. The same paired ATAC subsets and both-mate MAPQ30 selection are used for each assembly, with duplicate and organellar exclusion. Each reference supplies its own effective genome size; peak calling therefore also reflects a changed background model.

Scoring. Reciprocal conversion requires at least 95% aligned bases in each direction, unique placement and at least 95% round-trip span overlap. Jaccard uses eligible peak-base unions in the destination assembly. Spearman compares fragment counts on baseline intervals and their eligible projections; widths may differ.

Table 1. Input identities at the atac-arabidopsis-1m cap.

Library	Role / control	Raw templates
SRR4000468	biological replicate 1	1,000,000
SRR4000469	biological replicate 2	1,000,000

1 Eligible regions agree strongly across the two assemblies

At 1 million pairs, matched-region Spearman is 0.9979 and 0.9946 for 4,992 and 3,040 eligible TAIR10 peaks. At 5 million pairs, it is 0.9990 and 0.9977; these are existing recorded results, not newly rerun analyses.

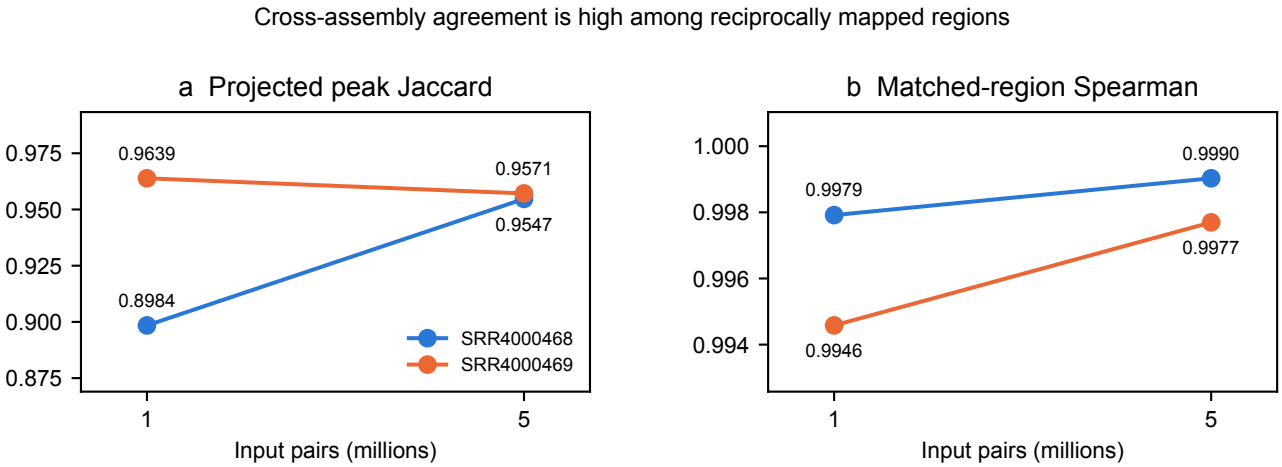


Figure 1. Reciprocal-coordinate agreement at two read depths.

2 Coordinate exclusions remain part of the result

Table 2. Peak eligibility before cross-assembly comparison.

Million pairs	Library	TAIR10 accepted / total	Col-CEN accepted / total
1	SRR4000468	4,992 / 4,997	5,586 / 5,596
1	SRR4000469	3,040 / 3,043	2,968 / 2,977
5	SRR4000468	14,788 / 14,805	15,312 / 15,343
5	SRR4000469	11,538 / 11,551	11,905 / 11,930

At the 1-million-pair cap, 8 TAIR10 and 19 Col-CEN peaks across the 2 libraries fail the reciprocal criteria. Excluded, ambiguous or unmapped intervals are not assigned to a convenient first placement.

Evidence revision. The public October 14 snapshot described the 5-million-pair mapping as pending. Its later completed assessment now has a matching SHA-256 completion receipt. The accompanying compact audit preserves the result, mapping, script and experiment hashes; it does not overwrite the historical report or distribute the local BAMs.

3 Reference distribution names do not guarantee new sequence

The A. lyrata audit reports identical case-insensitive sequence multisets for 695 contigs totaling 206,667,935 bases in the Ensembl v1.0 and Phytozome v2.1 distributions. All 695 contigs have unique full-sequence matches, while names and order differ.

Table 3. Reference preparation is distinct from a sequencing comparison.

Comparison	Completed evidence	Not established
A. lyrata distributions	695 exact contig matches	Newer sequence improves peaks
Maize AGPv4 / NAM5	1,003 forward / 2,413 reverse chains checked	Sequencing method superiority
Sorghum assemblies	Reference acquisition records	Validated peak concordance

Limits. Uniqueness is bounded by the supplied assembly alignments, not a proof that every repetitive placement is represented. Comparison within retained coordinates does not establish overall assembly accuracy, and neither reference has independent biological ground truth here.

4 Annotation changes TSS enrichment without changing reads

On identical Col-CEN BAMs, all-start TSS sets contain 46,558 Araport and 32,986 TAIR positions. For replicate SRR4000469, the score is 2.7517 with Araport versus 3.2096 with TAIR; the common-gene comparison retains 18,428 eligible starts per annotation.

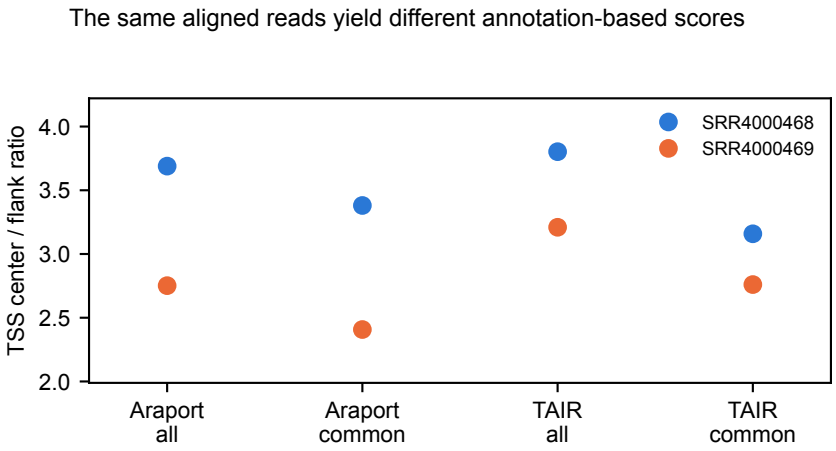


Figure 2. TSS enrichment with all-start and common-gene annotation sets.

Metric. Tn5 cut positions use zero-based start+4 and end-6. The enrichment ratio divides the center bin by mean outer 100-base flanks across full 2,000-base windows on each side. Of 18,430 common single-start genes, 2 edge positions lack full windows; common genes can have different TSS positions across annotations.

Open issues

- Higher TSS enrichment does not prove better annotation or library quality.
- Matched-coordinate agreement excludes precisely some difficult assembly regions.
- The broader species sequencing panel is not established by reference acquisition alone.

Recommendation

We would record sequence identity and annotation suitability as separate decisions.

1. Choose the assembly and gene models required by the intended downstream task.
2. Retain coordinate exclusions and compare independent regulatory evidence.

Appendix: Methods

We summarize recorded results without rerunning biological processing. The main Arabidopsis matrix uses seeded nested raw-template subsets (seed 60141), three fresh technical repetitions at the 1-million cap and one at the 5-million cap. Actual counts replace requested caps for short libraries. DAP retains primary mapped reads including MAPQ 0; ATAC selects proper primary pairs with both mates MAPQ at least 30, excludes MAPQ 255, organelles and marked duplicates. Biological identity is preserved. Source files are SHA-256 checked against the accompanying catalog; evidence snapshot revision ed1c50b8e4b5 includes the October 13 to 16 records. Completed local reference assessments are separately identified in that catalog.